



Invasive species in China – an overview

XIE YAN^{1,*}, LI ZHENYU², WILLIAM P. GREGG³ and LI DIANMO¹

¹*Institute of Zoology, Chinese Academy of Sciences, 19 Zhongguancun Lu, Haidian, Beijing 100080, China;* ²*Institute of Botany, Laboratory of Systematic and Evolutionary Botany, Chinese Academy of Sciences, 20 Nanxincun, Xiangshan, Beijing 100093, China;* ³*U.S. Geological Survey, 12201 Sunrise Valley Dr, MS300 Reston VA 20192 Washington DC USA;* **Author for correspondence*
(fax: (8610) 6264 7675; e-mail: xieyan@public3.bta.net.cn)

Received 6 June 2000; accepted in revised form 25 September 2000

Abstract. China is a vast country with rich biodiversity, which makes China especially vulnerable to invasive species. It has a long history of introduction of non-native species, especially those with perceived beneficial impacts. Its rapid economic development, including an explosive growth in international trade and transportation, has increased the potential for new introductions. Currently, alien species are widespread in the country, occur in many ecosystems, represent most major taxonomic groups, and are introduced unintentionally as well as intentionally for cultivation. The paper lists various cases of invasive species which have caused significant threats or damages to local natural or artificial ecosystems, and indicates that two example industries (fresh water fisheries and lawn grasses) have brought or tend to bring in many invasive species and hence have caused or will cause changes and loss of biodiversity in local ecosystems. Based on these studies, it is suggested that China combat the problem through enhancing awareness, development of a database on invasive species, strengthening international co-operation, preparing case studies and introducing the necessary legislation, regulations and monitoring.

Key words: China, fresh water fisheries, introduction, invasive species, lawn grasses

Introduction

China (Taiwan, Hong Kong and Macao are considered as a part of China) is the world's third largest country and one of the richest in terms of biodiversity. Its vast territory stretches 5200 km from east to west. It spans 50 degrees of latitude, and covers five climatic zones: cold-temperate, temperate, warm-temperate, subtropical, and tropical (Wang Sung et al. 1997). A wide range of habitats and environmental conditions makes China especially vulnerable to the establishment of invasive species of foreign origin. Potential invasive alien species from most areas of the world may find suitable habitat somewhere in China.

China has a long history of introduction of non-native species, especially species proven to be productive elsewhere and offering potential economic benefits to China. Early introductions were associated with immigration and trade among different regions. In the 4th century B.C., 'the Shu – Yuan du' ('Sichuan – India') trade route, including a network of local routes, linked Chengdu in Sichuan, via Tengchong in Yunnan and the north of Burma, to India. Tamarind (*Tamarindus indica*), originally

from Africa, was widely introduced and cultivated in the region and has naturalized in the valley of Yuanjiang, Yunnan. During the Han Dynasty (206 B.C.–220 A.D.) a ‘Silk Road’ connected the Weihe River in northwestern China, via countries in central Asia, with eastern Turkey. In 126 B.C., Zhang Qian and his assistant – messengers of Han Dynasty – brought back seeds of economic plants to China from central Asia, including grapevine (*Vitis vinifera*), alfalfa (*Medicago sativa*), common pomegranate (*Punica granatum*), and safflower (*Carthamus tinctorius*). All except grapevine have escaped to the wild in western China. The Tang Dynasty (618–907 A.D.) was particularly involved with the importation of exotic goods from almost every nation of Asia. Schafer (1963) summarizes and portrays the vast extent and breadth of these exotic imports, in particular wildlife, plants, foods and medical drugs. During the Song Dynasty (960–1279 A.D.) Quanzhou and Guangzhou became main ports connecting China with southeast Asian countries. Barbados aloe (*Aloe barbadensis* or *A. vera* var. *chinensis*), originally from northeast Africa, was first recorded as an imported medicine in the book of *Bencao Shiyi* (*A Supplement to Medicinal Herbs*) written by Chen Cangqi in 739 A.D. The plant was described as a free-living population between during 969 and 975 A.D. and has become widely naturalized in the coastal region in south China. In 1645, western Europeans arriving in India and southeast Asia by the ‘Gama Sea Route’ introduced species newly collected from America. Chinese living in the region subsequently brought back many of these species to China, including economic plants such as sweet potato (*Ipomoea batatas*), tobacco (*Nicotiana tabacum*) and blue passionflower (*Passiflora coerulea*). They also unintentionally introduced invasive weeds such as Spanish needles (*Bidens bipinnata*). During the Dutch occupation of Taiwan (1624–1662), some American species, such as horse tamarind (*Leucaena leucocephala*), popinac (*Acacia farnesiana*), and queen-of-the-night (*Hylocereus undatus*) and others, were introduced into Taiwan and have naturalized locally. After the Opium War in 1842, the ports of Hong Kong, Guangzhou, Xiamen, Shanghai, Qingdao, Yantai and Dalian became major pathways for the introduction of exotic weeds to China. For example, flax-leaf fleabane (*Conyza bonariensis*) was first reported in Hong Kong in 1857, horse-weed (*Conyza canadensis*) in Yantai, Shangdong, in 1862, and daisy fleabane (*Erigeron annuus*) in Shanghai in 1886. These species subsequently became invasive in wild lands. Other plants were introduced through a growing variety of pathways. For example, gorse (*Ulex europaeus*) from Europe was introduced by a French missionary to a church in Chengkou, Sichuan, and then escaped to the wild (Wei Zhi and Wei Siqu 1998).

China’s rapid economic development in the 20th century, including explosive growth in trade and transportation systems, is increasing the pathways for the introduction and spread of invasive species among regions within China and the introduction of new invasive species to China from other countries. Since many invasive species have long lag times from initial establishment until the appearance of a full-blown invasion, the full effects of the invaders arriving in China may not be felt until well into the 21st century. In the United States, the globalization of travel and trade

has facilitated introductions of invasive species from the early days of the Industrial Revolution in the 19th century. Many species first introduced decades ago have only recently begun to spread rapidly in ecosystems. There has been an upward trend in the establishment of non-indigenous species, and large numbers of invasive species have been documented (Randall and Marinelli 1996; Vitousek et al. 1997; Nico and Fuller 1999). In view of the ecogeographic similarities between the United States and China, the US situation may be a bellwether of future trends in China as China's world trade and domestic development continue to expand.

For many years, government departments of agriculture, forestry, and animal husbandry, as well as customs authorities in China, have paid great attention to the potential harm of a small number of alien species, leading to the quarantine of alien diseases and pests. However, it was only a few years ago that the concept of invasive species was introduced into China and the potential threats to China's natural heritage are not yet widely recognized. This contrasts with the situation in a number of growing countries where invasive species have become an important environmental issue. There are already many studies and publications, mostly from developed countries, on alien mammals, plants, weeds, fishes, molluscs, crustaceans, herpetofauna and wildlife diseases, including general discussions of pathways, vectors, the role of natural and human-caused disturbances (physical, chemical, climatic, etc.), and the economic and environmental impacts of invasions (Westbrooks 1998; Sandlund et al. 1999). The World Conservation Union (IUCN) has also established the Invasive Species Specialist Group (ISSG) to address problems of invasive species. The Convention on Biological Diversity (CBD), to which China and 177 other countries are Party, calls on governments to 'prevent the introduction of, control or eradicate those alien species which threaten ecosystems, habitats or species' (Article 8h). However in China, except for the attention given those species that have created great economic loss or have been listed as quarantine species by the Bureau of Animal and Plant Quarantine, little research has been conducted on invasive alien species and their impacts on natural ecosystems. There are no complete statistics or materials describing the status of alien species in China and their potential to create harm, and control measures directed at alien species are not well documented. This report, therefore, provides only a preliminary review of invasive species in the country. In our review of scientific literature and other documentation (Li Yanghan 1998), we have identified 380 species of vascular plants, belonging to 62 families and 210 genera, that have become invasive in China. The list includes species in natural ecosystems, agricultural lands, and other intensively managed areas. Descriptions are available in published literature cited for most of the species mentioned in the report.

In this report, we use the following criteria to define an invasive alien species that is: (1) has been introduced from its natural range into an area where it is not native by either intentional or unintentional human activity; (2) has established a self-reproducing population in a local natural or artificial ecosystem; and (3) has

caused obvious changes in a natural ecosystem or landscape, or has caused damages to a local natural or artificial ecosystem. This meets the implied definition of alien invasive species contained in the CBD. We also consider alien species in China to include species introduced outside their native ranges within China as well as non-native species introduced from other countries.

Alien species occur in each of China's 34 provinces, municipalities and autonomous regions. They occur widely in both urban and rural landscapes, and in protected areas. Alien plants have been reported everywhere, except in a few remote reserves in the Qinghai–Tibet Plateau, Hengduan Mountain, Xinjiang and Inner Mongolia. Alien species occur in almost every watershed and ecosystem, including forests, wetlands, grasslands, and croplands. They represent many taxonomic groups, including mammals, birds, reptiles, amphibians, fishes; arthropods and crustaceans; algae, ferns and seed plants; and fungi, viruses, bacteria, and other micro-organisms.

Unintentional introductions

Most documented alien species in China have been introduced unintentionally. As travel and trade have increased, travellers and transported goods have provided pathways for introducing more and more species as hitch-hikers on clothing and vehicles; in containers, packaging materials, shipping waste, and ships ballast; and as contaminants in shipments of agricultural seeds and other biological materials. Some of the most invasive, damaging and widespread unintentional introductions include the banana moth (*Opogona sacchari*), fall webworm (*Hyphantria cunea*), brown rat (*Rattus norvegicus*), ragweed (*Ambrosia* spp.), amaranth (*Amaranthus* spp.), goldenrod (*Solidago* spp.), crofton weed (*Eupatorium adenophorum*), and spinyflower alternanthera (*Alternanthera pungens*).

China's rapidly increasing trade is accelerating the introduction of alien species. According to government quarantine statistics, during the eighth five-year plan (1990–1995), 65 million tons of food were imported, mainly from the USA, Canada, Australia, the European Union, and Argentina. Also, cross border food trade has greatly expanded with Vietnam, Thailand, Myanmar, and Nepal. Because of the large number of trading partners and modes of transportation, the probability of bringing in alien species is high. From 1986 to 1990, nearly 200 species of weed seeds from 30 families and 100 genera were intercepted from 349 ships at the port in Shanghai. It is estimated that a 30,000 ton ship can bring 120 tons of weed seeds which, if widely disseminated, might cause great damage (Yin Lianping 1998). According to statistics of Li Yanghan (1998), weed seeds of 547 species and 5 varieties, belonging to 49 families, had been intercepted in 12 ports including Dalian, Qingdao, Shanghai, Zhangjiagang, Nanjing, and Guangzhou. They were brought in from 30 countries and are considered as contaminants in imported food, forage, cotton, wool, grass-sward and other economic plant seeds. About 170 of these alien weeds are invading new areas, including wild lands.

Intentional introductions

Institutions often have introduced species from other countries and from one region of China to another for economic development, recreation uses, or environmental improvements such as erosion control, landscaping, highway beautification, and creation of wildlife habitat. Many of these species have been introduced by institutions involved in agriculture, animal husbandry, forestry, horticulture, and aquaculture, and species specific centers for farming and breeding. Many of these species are mainstays of China's economy.

The number of intentionally introduced species in China has increased rapidly in recent years. For example, except for *Zoysia materlla*, all lawn grass seed is imported, with 2000 tons imported in 1997 alone (Xu Xiaogang et al. 1999). In 1990 and 1991, China imported more than 40 species from Taiwan as breeding materials for developing food, oil, melon, fruit and vegetable production (Ye Qixian and Hua Li 1997). There have been numerous introductions for aquaculture and specialized farming. These include the freshwater prawn (*Macrobrachium rosenbergii*), red claw (*Cherax quadricianalus*), salmon (*Oncorhynchus mykiss*), tilapia (*Tilapia* sp.), European eel (*Anguilla anguilla*), paddlefish (*Polyodon spathula*), colossoma (*Colossoma brachypomum*), striped bass (*Morone saxatilis*), large-mouth bass (*Micropterus salmoides*) and white bass (*Morone chrysops*) (Qiu Liming and Deng Guosheng 1994; Liao Guozhang 1998). No reports on the impact of many of these species in local aquatic ecosystems are available.

China at risk

A growing number of invasive alien species are causing significant economic and environmental impacts in China. These species include microbial pathogens, nematodes, insects, crustaceans, molluscs, fish, birds, mammals, and vascular plants. Appendix 1 provides examples of invasive species from these major taxonomic groups that are causing particularly significant impacts. The examples include available information on country of origin, the year of introduction, the purpose of introduction, means of spread, area of spread and the damage caused.

Unintentional introductions

Unintentionally introduced diseases and pests have caused great economic loss in many sectors, such as agriculture, husbandry, forestry and horticulture. These losses have drawn the attention of government agencies responsible for customs and quarantine. Particularly serious species in agriculture include black spot of sweet potato (*Ceratocystis fimbriata*), fall webworm (*Hyphantria cunea*), vegetable leaf miner (*Liriomyza sarivae*), and ragweed (*Ambrosia* spp.); in animal husbandry,

crofton weed (*Eupatorium adenophorum*); in forestry, fall webworm and alligator weed (*Alternanthera pungens*); and in horticulture, banana moth (*Opogona sacchari*).

Intentional introductions

A small percentage of intentionally introduced species have escaped cultivation and become invasive. These species can cause economic or environmental harm that can offset the benefits from their cultivation.

The four popular fish farming species, black carp (*Mylopharyngodon piceus*), silver carp (*Hypophthalmichthys molitrix*), bighead (*Aristichthys nobilis*), and grass carp (*Ctenopharyngodon idella*), were widely distributed and bred in most rivers and lakes in China. They have been also introduced from their native ranges to other regions of China, and bighead and silver carp have been reported to cause serious disruption. Other intentional vertebrate introductions include nutria (*Myocastor coypus*) from South America to support the fur trade. Several introduced vascular plants also cause significant disruption, including water hyacinth (*Eichhornia crassipes*) and alligator weed (*Alternanthera philoxeroides*) (both originating in South America), and common cordgrass (*Spartina anglica*) from western Europe. These species were variously introduced for ornamental use, forage, improving water quality, soil improvement, and erosion control. All are highly invasive and outcompete native species.

Recently, some introduced water plants have escaped from aquaria. These include Washington grass or fanwort (*Cabomba caroliniana*), imported from the United States, which now occurs in rivers in Zhejiang Province (Ding Bingyang 2000); and Parrot's feather (*Myriophyllum aquaticum*), from South America, which is naturalized in Taiwan (Li Zhenyu and Hsieh Changfu 1996). Although neither species is known to produce seed in its introduced range in China, both could quickly expand their range by asexual reproduction.

The special case of alien species in fresh water fisheries

Invasive species have changed local ecosystems by modifying species composition, population structure, and food chains. Watersheds in Yunnan Province are a good example (Chen Yingrui et al. 1998). Yunnan is one of the richest provinces in China in terms of biodiversity. The province has 432 documented fresh water fish species, or 42.2% of total number in China. The six river systems each has its special fish fauna. These river systems have only 3–10 species in common. Extreme isolation has facilitated evolution of many endemic fish populations in more than 10 lakes associated with these river ecosystems. Fish species composition is quite different not only between lakes in different river systems, but also between different lakes in the same river system. For example, Xingyun Lake and Fuxian Lake, located in the same

river system and only separated by one river, have only 11 species in common, while Fuxian Lake has 25 fish species and Xingyun Lake 14. One third of Yunnan's 432 fish species are either threatened or extinct due to the cumulative effects of over fishing, dam construction, water pollution, vegetation destruction, and land reclamation from lakes. In recent years, the spread of invasive alien species has been correlated with local extinctions and population reductions of the remaining native fish. For example, by the early 1970s, more than 30 alien fish species had been introduced into Yunnan's Dianchi Lake. Correspondingly, the number of indigenous species declined from 25 in the 1940s, to 15 in 1978 and 8 in 1982. According to a 1997 study, although several widely distributed native species such as the weatherfish (*Misgurnus anguillicaudatus*), Asian swamp eel (*Monopterus albus*), and crucian carp (*Carassius auratus*) still occur in the lake, only two endemic local species, *Yunnanilus plenrotaenia* (a kind of river loach, Balitoridae) and *Sinocyclocheilus grahami grahami* (a kind of barb), remain, and in small numbers in a greatly reduced range.

Significant economic damage has been caused by the introduction of alien species in freshwater fisheries. However, this damage and changes to local ecosystems have not drawn the attention of the public. Although there have been many comprehensive scientific surveys on fish conducted in large natural watersheds including lakes, rivers, and estuaries, fishery research has focused only on the biology and status of economically important fish. Very little research has been conducted on the impact of alien species (including species introduced for aquaculture) on species populations, biological diversity, and trophic interactions in fresh water aquatic ecosystems (Liao Guozhang 1998).

Effects on lake ecosystems associated with fishery management can be divided into three periods. Prior to the 1950s, the focus was primarily on managing a natural fishery – a strategy which in the present day applies only to some middle-to-large-sized lakes, such as Dongting Lake in Hunan and Poyang Lake in Jiangxi. Since the 1950s, emphasis shifted toward the cultivation and release of popular species, such as silver carp (*Hypophthalmichthys molitrix*), bighead (*Aristichthys nobilis*), and grass carp (*Ctenopharyngodon idella*) in small-to-middle-sized lakes. In these cases, algae replace aquatic macrophytes as the main primary producers, and the natural biological productivity of the lakes is fully consumed by grazing herbivores which outcompete native species. The aquaculture production in such systems generally exceeds the productivity of the natural fishery. These are the so-called 'algae type' lakes. Beginning in the 1980s, more attention was given to protecting and restoring native fisheries, while at the same time exploiting water resources and developing aquaculture. In such management systems, fish raised in aquaculture are kept within fish fences to allow aquatic macrophytes to grow. These are the so-called 'grass type' lakes (Zhang Guohua et al. 1997).

Donghu Lake in Wuhan, Hunan Province, is a good case study. In order to provide more forage for bighead and silver carp, and make their harvest easier, herbivorous fishes were carefully released. Due to a lack of aquatic macrophytes, the abundance of

zooplankton was greatly reduced. Algae increased dramatically. pH values became higher and the number of acid-favoring species was reduced. The biodiversity dropped, the number and quantity of large zooplankton was reduced, and small-sized forms developed. According to investigations before 1972, there were 67 fish species in Donghu Lake, of which nearly 20 had economic value, such as common carp (*Cyprinus carpio*), crucian carp (*Carassius auratus*), yellowcheek carp (*Elopichthys bambusa*), *Ochetobius elongatus* (Cyprinidae), bighead, silver carp, grass carp, black carp (*Mylopharyngodon piceus*), white bream (*Parabramis pekinensis*), redfin culter (*Culter erythropterus*), *Xenocypris* (Cyprinidae) and others. After aquaculture, the rate of growth of cultivated species increased until the growth rates for bighead and silver carp stabilized at around 98% of the total fish productivity. The structure of the fish community in Donghu Lake was changed completely (Zhang Guohua et al. 1997).

This kind of aquaculture system, first developed in eastern lakes, has been introduced into lakes in other parts of China, such as Keluke Lake, Dalian Lake, Genggahai Lake and Alarhai Lake in the Qinghai-Tibet area; Daihai Lake in the Mongolian-Uygur area; Qilu Lake, Dianchi Lake, Erhai Lake, Xingyun Lake and Lugu Lake in the Yunnan-Guizhou area. The introduced species compete with indigenous species for food and habitat resources, leading to greatly reduced numbers of local species, such as barbless carp (*Cyprinus pellegrini*) in Qilu Lake and Xingyun Lake, and even the local extinction of some species such as schizothoracins (*Schizothorax*) in Lugu Lake and Erhai Lake (Zhang Guohua et al. 1997).

Many alien fish species, such as European eel, large-mouth bass, paddlefish, colossoma, striped bass and white bass, have been introduced to south China and cultivated over a wide area. We do not yet know what impact such introduced carnivores, especially those like large-mouth bass, striped bass and white bass, that feed on small-sized fishes, have had on local fish populations in natural watersheds.

Lawn grasses – an emerging problem?

There is no evidence that introduced lawn grass species have invaded natural ecosystems in China. However, grasses are among the most easily spread of alien species. The risks to natural ecosystems are thus a matter of concern because the turf grass industry is developing rapidly in China. Large areas of lawn are being planted for golf courses, soccer fields, parks, and other purposes. The growing demand for turf grass seed has stimulated the development of a major industry focused on introduction and breeding of varieties from other countries. Except for *Zoysia materlla*, seeds of all other species of turf grasses are imported – more than 2000 tons in 1997. By 1990, 114 different kinds of cold-season grass seeds had been introduced from the United States. This has occurred in spite of the fact that China's diverse biogeographic regions provide abundant germplasm resources for native grasses. Through a research program spanning nearly 20 years, the Chinese Academy of Agriculture Sciences (CAAS) has collected

and preserved more than 1000 grass species suitable for use in lawns. Recent investigations on wild grass germplasm resources have demonstrated positive results. For example, more than 20 warm-season lawn grasses are widely used in the world. Except for a few species (e.g., *Paspalum notatum* and *Buchloe dactyloides*) many are naturally distributed in China. Many of the 136 widely used cold-season lawn grasses can be found in the northern and southwestern parts of China (Xu Xiaogang et al. 1999). However, China ignores the rich resource of native grass species, large numbers of grass seeds have been and still being introduced into cities from other countries, and their potential effects of these species on surrounding ecosystems are unknown. To date, there seems to be little concern about the potential harm that could result from the spread of some of these species into natural ecosystems.

Biological control agents

A number of species have been intentionally introduced into China for biological control of diseases, weeds and insect pests. In the early 1900s, Taiwan introduced vedalia beetle (*Rodolia cardinalis*) from California and Hawaii to control cottony cushion scale (*Icerya purchasi*). This is the first record of the successful introduction of a natural enemy in China. The number of successful introductions has increased since the 1950s. These include a monophagous parasitoid (*Aphelinus mali* – Hymenoptera: Aphelinidae) to control woolly apple aphid (*Eriosoma lanigerum*), and Australian ladybird (*Cryptolaemus montrouzieri*) to control mealy bugs (*Pseudococcus* spp.). Between 1979 and 1985, 182 biological control agents were introduced, excluding those in Taiwan and by the Ministry of Forestry. Several have been highly successful, including a parasitic wasp (*Encarsia formosa*) to control the greenhouse whitefly (*Trialeurodes vaporariorum*), western predatory mite (*Metaseiulus occidentalis*) and phytoseiid mites (*Phytoseiulus persimilis*). Additional progress was made in the introduction of some other species such as *Zygogramma suturalis* (Coleoptera: Chrysomelidae) to control ragweed (*Ambrosia* spp.) and water hyacinth weevil (*Nechetina* spp.) to control water hyacinth. Between 1986 and 1995, there were 101 introductions (excluding Taiwan). Among these was the successful introduction of a parasitic wasp (*Coccobius azumai*) from Japan to Guangdong in 1989 to control pine scale (Pan Wuyao et al. 1993). According to statistics of the Guangdong Forest Bureau, the use of biological control technology has saved about US\$ 5,000,000 each year in controlling pine scale (Lu Qingguang 1997a,b). Although China's experience with biological control has been positive, various countries have documented cases in which introduced biological control agents have themselves become nuisance species. For example, weevils introduced into the United States to control a Eurasian thistle are now threatening indigenous North American thistles (Simberloff 1997). These cases remind us that assessments of the potential harmfulness of these species to non-target species prior to release and careful monitoring of the impacts in local ecosystems following release are essential.

Public awareness and education – a critical need

In China, the general public is not aware of the harm now being caused by alien species and their potential future risks to local ecosystems and China's rich biological heritage. Currently no Chinese publications or guidelines on invasive species are available in China. The need for education extends to many sectors of Chinese society from policy makers, resource managers and researchers to workers, farmers, and the general public.

People are aware that diseases and pests cause economic loss, but have not realized that alien species are damaging ecosystems and outcompeting native species. Alien species that cause these impacts but do not directly reduce economic production have received little attention. There is a general lack of understanding of the impacts of invasive species on ecosystem functions. For example, farm-raised nutrias and confiscated live wildlife in cross-border trade are released into local ecosystems without regard to their potential effects on these ecosystems. The banana moth is a significant pest of fragrant dracaena (*Dracena fragrans*), an important ornamental species in China. A 1997 survey found that severely infested fragrans dracaena plants were discarded in a corner of a flower nursery. Pest densities of 100 per square meter were reported. Known host plants for the banana moth include more than 50 species from 24 families, including many crops such as banana, sugarcane, corn, sweet potato and ornamental plants. Thus, the damage could be serious if the sale and improper disposal of the infested stock enabled the pest to spread from nursery stock to crops and natural ecosystems (Chen Guifang et al. 1998).

The primary purpose of introductions is to increase economic returns, and assessments of the potential dangers to local ecosystems, native species, and economic uses of resources have rarely been made before species are introduced. Such impacts are also rarely considered in natural resource management. For example, some nature reserves are planning to begin reforestation using alien species, such as the proposed use of species of *Rhizophora* from Bangladesh to reforest mangroves in the Dongzaigang Nature Reserve (MacKinnon, pers. comm.). There is also a serious lack of research and long-term monitoring on these kinds of impacts to help support scientifically based assessments.

A balanced approach to the threats requires awareness of how different people benefit and are harmed by invasive species. Many invasive species – for example, some commercially important fish and ornamental plants – have strong constituencies who favor these species over native species.

Action now

The alien invasive species issue should be put on China's national agenda immediately. Action must be taken now. Inaction will result in substantial economic loss and damage to China's endemic ecosystems and rich biological heritage. Here are some suggestions for action.

Compile educational materials

Collect and compile materials, including concepts, dangers, impacts, prevention and control techniques, management experiences (both successes and failures) and case studies relating to invasive species. Booklets, posters, exhibits, and other materials should be attractively designed for the general public with copious illustrations and photos. Information should be distributed through the mails, the Internet, bookstores, and various media, such as newspapers, radio, and television.

Develop databases and information systems on invasive species

Collect available publications and data on known and potential invasive species, including taxonomy, native and introduced ranges, biology, physiology, and impacts; pathways for introduction and spread; prevention and control methods and other information. Enter the information into databases, develop related data documentation standards and search tools, and make information accessible through web sites, CD-ROMs or other suitable channels to meet the needs of potential users.

Strengthen international co-operation

It is particularly important to strengthen international communication and co-operation in sharing, linking and integrating invasive species databases and information systems, and in conducting research to support effective prevention and control strategies. International cooperation is extremely important for several reasons:

Countries seeking to address threats from a particular invasive species need information on the taxonomy, ecology, natural enemies, and management of the species from countries where the species is native. The successful introduction of a parasitic wasp (*Coccobius azumai*) from Japan, the country of origin for pine scale, to control the spread of this pest in China, is a good example of the value of looking for a natural enemy within the invaders native range (Chen Yongge and Gu Dexiang 1998).

Knowledge gained on an invasive species and its control in one country can be valuable to other countries in making decisions on prevention and control of the same species. For example, information on water hyacinth invasions in North America, Asia, Australia and Africa can provide important insights for developing strategies for prevention and control of the same species in China. In another example, Taiwan introduced vedalia beetle (*Rodolia cardinalis*) from California and Hawaii to control cottony cushion scale (*Icerya purchasi*) early in this century (Lu Qingguang 1997a,b). The success of this introduction was partly due to lessons learned from introducing the same species in the United States.

Just as China has suffered from invasive species introduced from countries where they are native, so are many countries plagued by invaders that are native to China. For instance, grass carp, a widely distributed and raised native species in China,

has been reported from 45 states in the United States where the species competes with for food with invertebrates (e.g., crayfish) and other fishes, causes significant changes in macrophyte, phytoplankton and invertebrate communities, interferes with the reproduction of other fishes, decreases available refugia for other fishes, and so on (Nico and Fuller 1999). Chinese tallow tree (*Sapium sebiferum*), first introduced into the US in 1776, has become an aggressive invader which is decreasing native plant diversity along the Gulf or Mexico and Atlantic Coasts from Texas to North Carolina (Randall and Marinelli 1996). A more recent example is the Asian longhorned beetle (*Anoplophora glabripennis*), native to China, which has killed hardwood trees in local infestations in the United States. The beetle attacks many hardwoods, including Norway, sugar, silver, and red maple; horsechestnut; poplar; willow; elm; mulberry; and black locust. The insects bore into the trunk of the tree and lay their eggs inside. Because the insects spend all but the summer months inside the tree, it is virtually impossible to eradicate them with insecticides. Research has not yet produced a trap specific to this pest. Presently, the only effective treatment for eliminating the Asian longhorned beetle once it invades an area is to destroy all infested trees. Suppressing a 1996 infestation in New York City and Amityville, New York, neighborhoods cost the State and Federal Governments more than \$ 4 million. To date, the Asian longhorned beetle has been found at 26 scattered warehouse and residential sites in 14 States around the country. The US Animal and Plant Health Inspection Services pest risk analysis has determined that the species is hitchhiking into the United States in solid wood packing materials, such as pallets and crates, from China. (PPQ 1998). Many more cases could be listed.

Because China and other countries have similar environments, they often share concerns about alien species that are potentially invasive in these environments. In the USA, cornflower (*Centaurea cyanus*) is particularly invasive in native grasslands and prairies, especially in the northwest – habitats that are becoming increasingly rare. Cornflower is a close relative of the knapweed and starthistle, which are serious pests in crop lands and range lands as well as natural areas. Once established cornflower produces abundant viable seed, and populations increase quickly. Many grassland preserves are heavily infested with this attractive nuisance species (Randall and Marinelli 1996). In China, although there is no report of its dangers, beautiful flowers can be seen in large numbers in flower markets and nurseries in many cities in China. Nobody knows if the species has escaped to the wild, established free-living populations or caused damage. However, information from other countries should encourage us to study and monitor this species.

Conduct case study

It will be necessary to study the status and impacts of invasive species (especially intentionally introduced species) in several regions, in order to understand their status in China. These studies will improve public awareness of the impact of alien

species to the national economy and ecosystems. They should also draw attention to the importance of preventing and controlling the introduction and spread of invasive species by providing managers, farmers, other stakeholders and the general public with concrete examples and scientific analyses.

First, take a middle-to-large-sized city and its surrounding area as a case study. Make an inventory of alien trees, shrubs, weeds, and herbaceous species; investigate surrounding natural ecosystems to document alien species that have invaded natural communities; conduct studies to understand and predict their impact on populations of native species, species composition, and ecological relationships, and select species for long-term monitoring.

In addition, select a middle-to-large-sized watershed as a case study, and study the effects of alien species effects on local species composition, population structure, quantities and food chains in the watershed. Integrate invasive species into a long-term monitoring program for documenting changes in the ecology and species structure of the watershed. Develop recommendations on strategies for reducing the effects of invasive species and other ecosystem stressors.

Introduce legislation

China already has one law and various regulations regarding quarantine. The People's Republic of China (PRC) Animal and Plant Quarantine Law took effect in 1992 and the related Implementing Regulations for this law took effect in 1997. The PRC Domestic Animals Epidemic Prevention Regulation and PRC Plant Quarantine Regulations were issued in 1983 and revised in 1992. Associated with these regulations are lists of epidemic and parasitic diseases of animals imported from other countries into the PRC, a list of animals, animal products and other quarantine goods which are forbidden to be carried or mailed into the PRC, a species list of dangerous diseases, insects and weeds; a named list of prohibited goods for importation was issued by Ministry of Agriculture in 1993 (Yu Dahai and Cui Yanlin 1997; Liu Yuanzhi et al. 1998). However, all these laws and regulations are for quarantine of diseases, pests and weeds. Laws or regulations do not presently exist to prevent damage to local ecosystems and native species from invasive species. Such laws are needed and should be promoted. For example, a possible approach would be to prohibit large-scale use of non-indigenous species until small-scale trials of several generations show no signs of threats to local ecosystems. Legislation is also needed to restrict use of alien species in protected areas, and to encourage the use of domestic species in reforestation programmes.

Acknowledgements

The study has been supported by Biodiversity Working Group of the China Council for International Co-operation on Environment and Development (BWG/CCICED)

and the National Science Foundation of China (No. 39630030, 39870056). Many experts have made great contributions to the report. Dr Andrew Smith kindly edited the main report. Dr John Fellowes provided information in Hong Kong; Dr Pam Fuller, Dr Jacoby Carter and Dr Phillip Thomas provided information from the USA; Dr John MacKinnon provided information on birds and Prof Chen Deniu provided information on molluscs. Drs Wang Sung, John MacKinnon, Peter Schei and Jacoby Carter made excellent comments on the paper. Their contributions are gratefully acknowledged.

Appendix I. Cases of invasive species in china

Mammals

Nutria (Myocastor coypus)

Nutria is an herbivorous rodent native to Argentina, Chile and Uruguay. It was first introduced from the former USSR to the Northeast China Zoo for display. Since 1986, nutria have been bred and raised in large numbers in many areas, especially in southern China, because of its heavy, soft, fine fur. However, a market for nutria failed to develop. Factors included the high cost of young animals, poor fur quality, and unpalatable meat. By the mid-1990s, nutria farming was largely abandoned. Without management and control, nutrias were released or escaped. Many subsequently established wild populations, which are causing increasing damage to fruit trees and crops. The species is now regarded as a locally significant pest in orchards and crop lands (Xu Ruiqiu et al. 1997).

Muskrat (Ondatra zibethicus)

Muskrat was introduced to the former USSR from North America in 1927. By the 1950s, it was being widely raised in more than 500 farms. Muskrats escaped in Xinjiang from these farms due to poor management and their populations expanded quickly along the Ili River, Tarkent River and Erjisi River into China. In 1955 and 1957, populations were reported from Huma and Xingkai Lake in Heilongjiang Province. Muskrats make holes in banks and damage water supply and irrigation works, causing significant economic losses in arid areas. Since 1958, muskrat has been introduced to North China, Inner Mongolia, Guizhou and other provinces for the fur trade (Shou Zhenhuang 1962; Cheng Dasheng 1986).

Brown or Norway rat (Rattus norvegicus) and Sladen's rat (Rattus tanezumi)

Unlike muskrat from North America and red squirrel (*Sciurus vulgaris esalbidus*), which were artificially released from the former USSR around the early 1950s, brown rat and Sladen's rat were brought into Xinjiang Autonomous Region from central China as hitchhikers through the railways. The first brown rat was caught in Xinjiang in 1975, and wild populations are now well established in Turpan prefecture and Urumqi City in Xinjiang. Brown rat populations are expanding north and south in Xinjiang along railways, as well as to areas of agriculture and animal husbandry along roads. Survey data indicate a stable population structure, with breeding ages becoming a little earlier and body size getting larger. In the croplands of Miquan county, the brown rat is outcompeting other species of rats and becoming the predominant species (Zhang Daming et al. 1993).

Birds

Sulphur-crested Cockatoo (Cacatua sulphurea)

Native to Indonesia, the species has been a significant pest in Hong Kong for many years where it destroys ornamental trees (J. MacKinnon, pers. comm.; Reels 1998).

Rainbow lorikeet (*Trichoglossus haematodus*)

This parrot was first introduced from eastern Indonesia to Australia and later to Hong Kong where it is now quite common. The species damages ornamental trees and raids fruit groves. (J. MacKinnon, pers. comm.; Reels 1998).

Canada goose (*Anser canadensis*)

This large goose, native to North America, is spreading in Hebei where it is frequently reported in croplands, especially potato fields (J. MacKinnon, pers. comm.).

*Fishes**Bighead* (*Aristichthys nobilis*)

Bighead is one of the most popular farming fish in China and is widely distributed throughout China. However, in the northeast and the west of China, it was brought in artificially (Chen Yiyu et al. 1998). Bigheads large mouth and long and dense gill raker make it a highly effective filter-feeder, which outcompetes native species. For example, in 1958, the total fish population of Qilu Lake, Yunnan Province, consisted of about 50% barbless carp (*Cyprinus pellegrini*). Barbless carp feeds on planktonic animals. With a smaller mouth and a short, low-density gill raker, the species competes poorly with the bighead, which utilizes similar food resources. Following introduction of the bighead, barbless carp populations have declined significantly in Qilu Lake, as well as in Xingyun Lake, Yunnan Province, where measures to augment populations of barbless carp were initiated in 1988. Hatchery-reared barbless carp are raised in Huanggu Reservoir, and 150,000–200,000 fries are released to Xingyun Lake every year (Zhang Guohua et al. 1997; Chen Yingrui et al. 1998).

Gobies (Gobiidae) and *Topmouth Gudgeon* (*Pseudorasbora parva*)

Several species were brought unintentionally to lakes in the Yunnan-Guizhou area, including Erhai Lake, from the southern and southeastern parts of China, along with the introduction of four principal fish-farming species in the 1950s and 1960s. With very few carnivorous fish in the lakes, populations of these small fish expanded rapidly and soon became the predominant species. They not only outcompeted indigenous species for food and habitat, but also ate their eggs, which further reduced native populations and even eliminated some of them. For example, three native species of Schizothoracin (*Schizothorax*) developed from a common ancestor in Lugu Lake. All are believed to have become extinct due to egg predation from introduced small-sized fishes, such as topmouth gudgeon, and disturbance of their breeding area by introduced carp (Chen Yingrui et al. 1998).

Mosquitofish (*Gambusia affinis*) and *Livebearers* (Poeciliidae)

Mosquitofish is originally from the United States and northern Mexico. It was introduced into Taiwan in 1911 and from there to mainland China in 1924 to annihilate mosquitoes, and subsequently bred naturally in many provinces. In Hong Kong, circumstantial evidence suggests that introduced mosquitofish and several livebearers have reduced native fish populations by preying on small fishes and by harassing and nipping the fins of larger ones. Some minnows, such as the endemic *Aphyocypris lini*, may have been eliminated through competition or predation. The once widely distributed native medaka (*Oryzias latipes*), is now found only in the remaining habitats not yet colonized by livebearers. It is also possible that the endemic Romers frog (*Philautus romeri*) may have been eliminated from some areas by mosquito fish predation on the small tadpoles (Dudgeon and Corlett 1994).

*Crustaceans**Crayfish* (*Procambarus clarkii*)

Crayfish is native to the southern USA and northern Mexico. The species is widely farmed in the Americas, Asia and Africa. Crayfish was introduced into China from Japan in the early 1940s. It was first bred in Nanjing in Jiangsu and Chuxian in Anhui, and then spread to north, central and south China where it has

developed vast populations in the wild. Unlike most aquaculture species, the female crayfish incubates her eggs, conveniently avoiding the need for, and cost of hatcheries. Once a pond has been stocked, crayfish usually establish self-maintaining populations. Restocking is normally not required except following outbreaks of disease. Crayfish are frequently raised in combination with crops, especially rice; where they feed on crop residues and the associated food web. Crayfish which escape the harvest survive by burrowing into levees or pond banks, providing the brood stock for the next season (Yao Gendi et al. 1995; Zheng Shengshun 1999).

Molluscs

Amazonian snail (Ampullaria gigas)

Originally from the Amazonian Basin, the Amazonian snail was introduced into Guangdong in 1981 by a Brazilian Chinese, and has been widely raised for food in the province since 1984. However, overproduction and the failure of a robust culinary market to develop have been factors in the releases of the species to the wild. The Amazonian snail is a voracious feeder, with high reproductive potential, which has adapted readily to many local environments. It spreads quickly and damages rice, vegetable and water crops. Since 1988, the species has caused significant economic losses in more than 37 counties encompassing 25,000 hm² in Guangdong Province. The Amazonian snail was introduced from Guangdong to Jinjiang in Fujian province in 1985, escaped to wild, and was causing significant economic damage by 1992. Invasions have also been reported from Xishuanbanna in Yunnan province (Liu Zhongli 1989; Cai Hanxiong et al. 1990; Zhao Guoshan et al. 1996).

Giant African snail (Achatina fulica)

Originally from East Africa, the giant African snail was first reported in China by Henkots in 1931 on the campus of Xiamen University in Fujian as a hitch-hiker on plants brought from Singapore by a Singaporean Chinese. The species has subsequently spread to Guangdong, Hainan, Guangxi and Yunnan Provinces, where, in many areas, it has become the main molluscan pest of crops and local ecosystems. Because of its value as a delicious food, the giant African snail was also intentionally introduced in these and other provinces, including Hong Kong and Macao in 1932 and Taiwan in 1933. In Yunnan, the species was also introduced from Viet Nam and Hainan for cultivation (Zhou Weichuan et al. 1989; Chen Deniu et al. 1990; Chen Deniu 1991).

Insects

Termites (Termitidae)

Termites are insect pests that feed on wood and fiber. They are widespread and cause significant economic damage in many countries. In the 1980s, Zhang Yingjun conducted extensive surveys and compiled available documentation from the literature on the introduction and spread of termites in China. These publications document the introductions of *Incisitermes*, Taiwan subterranean termite (*Coptotermes formosanus*), rubber termite (*Coptotermes curvignathus*), Ahmaditermes, *Nasutitermes* and others from shipments of wood and wood products into Nanjing in Jiangsu Province; Xianyang in Shanxi; Qingdao in Shandong; Wuhan in Hubei; Yantai and Penglai in Shandong; in Beijing; Xi'an in Shanxi; Jiang'an in Sichuan and Ninghai in Zhejiang. Various species have caused serious damage in some regions. For example, since western drywood termite (*Incisitermes minor*) was first introduced in Ninghai, Zhejiang from the USA in 1937, it has established breeding populations which have caused serious damage to structures in this region. Control programs, initiated in 1975, have prevented further spread. In 1993, the Ministry of Agriculture included the rubber termite in its Species List of Dangerous Diseases, Insects and Weeds and Name List of Goods Forbidden to Import (Liu Yuanzhi et al. 1998).

Pine scale (Hemiberlesia pitysophila)

Pine scale (Coccoidea, Diaspididae) is native to Japan and Taiwan. It was first reported in Huidong County and Bao'an County in Guangdong Province in the late 70s, where it destroyed masson pine forests. It spread rapidly, infesting 110,000 hm² in 1983, 310,000 hm² in 1986, and 400,000 hm² in 1987. By the end of 1990, pine scale infested an area of 718,000 hm² which included more than 130,000 hm² of dead

masson pines. In 1989, Guangdong successfully introduced *Coccobius azumai*, a parasite of pine scale from Japan as a biological control agent. According to the Guangdong Forest Bureau, use of the biological control agent has saved US\$ 6,000,000 annually due to the control of pine scale infestations (Xiao Gangrou 1992; Lu Qingguan 1997b; Chen Yongge and Gu Dexiang 1998).

Fall webworm, American white moth (Hyphantria cunea)

Fall webworm (Lepidoptera, Arctiidae) is native to North America and was introduced into Dandong and Xinjin in Liaoning Province in 1979. Since then, it has spread to Shandong and Shanxi, where it has caused serious damage in orchards, forests, and croplands. Fall webworm has been reported to infest more than 100 plant species in China. Within its introduced range, the species totally defoliates trees in orchards, in parks and recreation areas, and in horticultural plantings along boulevards, causing significant mortality of trees in both urban and rural areas (Li Yunchao et al. 1991; Lu Qingguan 1997b).

Banana moth (Opogona sacchari)

Fragrant dracaena (*Dracaena fragrans*) from Guinea was introduced as an ornamental plant to Guangdong Province in the early 1980s and was widely planted in the northern part of China, including Beijing, in the 1990s. In the middle 1990s, the first banana moth infestations in Asia were reported on fragrant dracaena in Beijing, where many trees were killed. The moth has a wide global distribution. It was reported from the Mascarene Islands in the Indian Ocean in 1856, and subsequently from many regions in Africa and Europe, and later from South America and the West Indies, finally invading Florida in the 1980s. Caterpillars are reported to feed on more than 50 species of host plants representing 24 families, including crops such as banana, sugarcane, corn and sweet potato and many horticulture plants, especially fragrant dracaena. Banana moth now occurs in many areas where fragrant dracaena is planted, and is regarded as a particularly serious threat in the southern provinces of Guangzhou, Fujian and Hainan. Until 1997, reports of this pest have been confined to ornamental plantings and nurseries. However, it is believed to pose a potential threat to natural ecosystems (Chen Guifang and Yang Jikun 1997a,b; Chen Guifang et al. 1998).

Loblolly pine mealybug (Oracella acuta)

The loblolly pine mealybug was introduced into Taishan in Guangdong Province in 1989 in a shipment of 70 loblolly pine (*Pinus elliottii*) scions from the state of Georgia in the USA. Since then, its range has expanded to include many counties in Guangdong Province. By July 1993, it had damaged 136,000 hm² of pine forest. It is estimated that the pest is currently spreading at the rate of 70,000 hm² a year. Loblolly pine mealybug infests many species of pines, including *Pinus taeda*, *P. elliottii*, *P. palustris*, *P. echina* and *P. virginiana*. During the past 30 years, China has introduced many pine species from the USA, among which *P. elliottii*, *P. taeda* and *P. caribaea* are widely planted. These species and many endemic pine species are now threatened by the pest (Pang Xiongfei et al. 1994; He Guofeng 1995).

Vegetable leaf miner (Liriomyza sativae)

The vegetable leaf miner destroys the cotyledons of many plant species, especially in the Cucurbitaceae and Papilionaceae families and in such genera as *Physalis*, *Lepidium* and *Ambrosia*. The pest is widely distributed in nearly 40 countries and regions in America, Australia, Africa and Asia. The species was first reported in China from Hainan, in December, 1993 and has since spread to 21 provinces, municipalities and autonomous regions covering more than 1,300,000 hm² where it causes significant crop losses. Parasitic bees, representing 4 families, 13 genera and 34 species have been found to feed on this pest, and successful biological control is expected (Wang Fuxiang 1997; Xu Zaifu and Zeng Ling 1998).

American rice water weevil (Lissorhoptrus oryzophilus)

Originally from North America, the American rice water weevil was introduced into Japan in 1976, and subsequently to South Korea and then to North Korea in 1988. It was first reported in China from Tangshan in Hebei Province in 1988 and subsequently from Tianjin, Shandong, Liaoning, Jilin, Zhejiang, Fujian and Taiwan (1990). By 1997, it had been reported from 54 counties in 8 provinces and damaged 310,000 hm² of rice fields (Li Xingyu 1997; Wei Hongjun 1997).

American cockroach (*Periplaneta americana*)

American cockroach is a rapidly invading insect originally from South America. It is now widely distributed throughout the world, including China where it was first reported from Taiwan in 1931. Higher densities are found in warmer regions in the south of Huaihe. The species contaminates food and is a vector for several human diseases. It is particularly common in wineries, bean factories and washrooms (Zhang Shimei and Zhao Yongxiang 1996).

German cockroach (*Blattella germanica*)

Originally from Europe, the German cockroach has been widely distributed in the most of counties in the world. It feeds on food, such as corn and Chinese sorghum, and is regarded as household pest. It was first recorded in northeast China in 1935 from which it had spread to provinces in eastern China by the early 1960s, and then to northwestern and southwestern regions in the 1980s (Zhang Shimei and Zhao Yongxiang 1996).

Woolly apple aphid (*Eriosoma lanigerum*)

Originally from North America, the woolly apple aphid was reported in London in 1787, mainland Europe in 1801, and Switzerland in 1870. It was introduced to Japan from North America in 1872, found in South Korea in 1926, and introduced to northeast China from Japan in 1929. In China, it was subsequently reported from in Shandong in 1934 and now has spread to Dalian, Suizhong, and Xingcheng in Liaoning Province; Qingdao and Yantai in Shandong; Kunming, Lufeng, and Zhaotong in Yunnan; and Lhasa in Tibet. Species in the apple genus are the main hosts for the pest, which congregates at leaf scars or axils and feeds on juice. Infested trees are weakened and may even die from lack of nutrition. The aphid has spread over long distances on transported seedlings, scions and fruit baskets, and locally by flying and on farming equipment (Feng Mingxiang 1992).

Grape root louse (*Phylloxera vitifolii*)

Originally from Eastern North America, the grape root louse was first introduced from France to Yantai in Shandong Province in 1892 and has been reported to cause economic losses in parts of Taiwan, Liaoning, Shandong and Shanxi. Species in the grape genus (*Vitis*) are the pest's only hosts. The grape root louse seriously damages roots, causing root rot which often kills the plants. In some grape varieties, leaves are also damaged. The pest spreads over long distances on transported seedlings and cuttings, and locally on farming tools, by wind or flowing water (Feng Mingxiang 1992).

*Nematodes**North American pinewood nematode* (*Bursaphelenchus xylophilus*)

The North American pinewood nematode is originated in North America and was firstly found in Nanjing, Jiangsu Province in early 80s. It is spread by longhorn beetles, whose larvae bore into pine wood. The nematodes can kill a pine tree within six months. During the 1980s, this pest nearly eliminated the extensive plantations of *Pinus massoniana*, which is native to the Hong Kong region. It has established many disease centers in Jiangsu, Anhui, Guangdong and Zhejiang, and is spreading to surrounding areas from these centers (Wang Qiongyu 1992; Dudgeon and Corlett 1994).

*Fungi**Black spot* (*Ceratocystis fimbriata*)

Black spot is a major disease of sweet potato caused by *Ceratocystis fimbriata*. The fungus was introduced from Japan in 1937 to Liaoning Province from which it rapidly expanded its infected area. By 1980, it was reported from 26 provinces, municipalities and autonomous regions. The disease reduces production by decomposing sweet potato seeds and killing young plants. The fungus also stimulates sweet potato to produce ipomeamarone and ipomeanine. These toxic biochemicals are known to cause headache in humans and asthma or even death in cattle (Lu Qingguan 1997b).

Wildlife diseases

Infectious pancreatic necrosis virus in trout (IPNV)

Infectious Pancreatic Necrosis (IPN) of trout was first recorded in 1940 in Canada and is now widespread in Europe, Asia and America. Many countries have reported IPN, including the USA, Canada, Italy, Japan, Korea, Norway, Scotland, and South Africa. In China, the virus occurs on both the mainland China and in Taiwan. The RNA virus belongs to the genus *Aquabirnavirus*, family Birnaviridae. It not only infects trout, but also many other fishes and aquatic animals. Aquatic animals in at least 20 families can be naturally infected by IPNV, including lamprey, cyclostomes, teleosts, and some crustaceans. The mortality for trout can be very high, and is believed to be much lower for other species. However, no reports are available on the effects of IPN on wild aquatic animals in China (Yu Dahai and Cui Yanlin 1997).

Plants

Mexican tea (*Chenopodium ambrosioides*)

Mexican tea is native to tropical America and is now widely distributed in tropical and sub-tropical areas of the world. It was first collected in Tamsui, Taipei in Taiwan in 1864. Since 1912, it has spread widely as a roadside weed in the Hong Kong region (Dunn and Tutchter 1912). Now, it is widely distributed in tropical and subtropical areas in China.

Alligator weed (*Alternanthera philoxeroides*)

Originally from Brazil, alligator-weed was introduced to Shanghai and eastern China in the 1940s. Since the 1950s, it has been widely introduced for pig forage in many areas of southern China from which it has escaped into the wild. A 1986 survey showed that the range of the species encompassed 13,344,000 hm² and that the species had become the major weed of vegetable gardens, sweet potato fields and citrus orchards (Wang Ren and Wang Yuan 1998).

Springflower alternanthera (*Alternanthera pungens*)

Native to Central America, springflower was recently introduced in coastal and open areas of Xiamen in Fujian Province and Changjiang in Hainan Province. Because its flowers have stinging bristles, the plant is regarded as a troublesome weed. A related species, *A. paronychioides*, from tropical America, is found in Wenchang in Hainan and Qi'ao Island in Guangdong Province (Wu Telin 1994) and in Changhua and Pingtung in Taiwan (Yang Yuenpo and Liu Hoyih 1996).

Amaranth (*Amaranthus* spp.)

Prostrate amaranth (*Amaranthus blitoides*) from North America was first reported from Liaoning in 1875 and subsequently from Beijing. A closely related species, tumbleweed (*A. ablbus*) was first recorded in 1935. The two non-native species have established a number of wild populations in north and northeast China. In addition, pigweed (*A. retroflexus*) and thorny amaranth (*A. spinosus*) from tropical America and green amaranth (*A. viridis*) from tropical Africa are also invasive weeds and are more widely distributed.

Cacti (*Cactaceae*)

Pest pear (*Opuntia stricta* var *dillenii*), originally from coastal areas of the Caribbean, was introduced to Taiwan by the Dutch in 1645 (Yang Gongyi 1984b). Since then, it has invaded large areas of coastal southern China where the species is often found on dry, stony soils. The species has sharp spines and is hard to eradicate. Indian fig (*O. ficus-indica*), a species from Mexico, was introduced into Taiwan in 1645. Since then, it has become wild in five provinces of southwestern China (including the southeast of Tibet) where it is widespread in dry, hot valleys. Another related invasive species is the prickly pear (*O. monacantha*), which is native to South America. In 1625, Liu Wenzheng wrote in the book of Yunnan Local Records that this species had already been introduced to Yunnan by then. Today, the species is widely naturalized in Yunnan, Guangxi, Guangdong, Fujian and Taiwan (Li Zhenyu 1999).

Common lantana (*Lantana camara*)

Originally from tropical America and introduced to Taiwan from Spain during the end of the Ming Dynasty (Yang Gonggi 1984a), common lantana has spread throughout southern China. This shrub, with its spiny stems and unpleasant odor, is a significant pest which has displaced native local plants.

Love apple (*Solanum aculeatissimum*)

Love apple is native to tropical areas of South America. Some naturalized specimens were collected in Guizhou at the end of 19th century, and mistaken for two new species (*S. bodinieri* and *S. cavaleriei*). The species may have been introduced earlier to Guizhou as an ornamental plant. The plant has spiny stems and leaves and has spread widely south of the Yangtze River where it is now a common weed. Its poisonous fruits are potentially lethal to foraging cattle. The genus *Solanum* includes 12 additional non-native species that are naturalized in China, including mountain tobacco (*S. erianthum*) and wild tomato (*S. torvum*). Except for the kangaroo apple (*S. laciniatum*) from Oceania, all were introduced from the Western Hemisphere.

Plantain (*Plantago* spp.)

Hoary plantain (*Plantago virginica*) was introduced from North America. It was first collected in Nanchang in Jiangxi Province in 1951 and now occurs in grassland and along roadsides and lake shores in the southern part of Jiangsu Province and in Anhui, Zhejiang, Jiangxi, Fujian and the north of Taiwan. It has been spreading rapidly due to its prolific seed production and adaptability. Bracted plantain (*P. aristata*) is another weed from North America, which was first found in Qingdao, Shandong Province in 1929. It has since become a widespread weed in Qingdao and now also occurs in Jiangsu Province where it grows on coasts, beaches, along roadsides, on hills and in grasslands (Li Zhenyu, in press).

Venus' looking-glass (*Triodanis*)

Venus' looking-glass (*Triodanis perfoliata*) is native to North and Central America and has become wild in South America and Asia. In China, it was first reported from Wuyi Mountain in 1974, and subsequently from various places in Fujian Province in the 1980s. It grows along streams, in grasslands and on hills at elevations from 180 to 1000 m. A related species, the small Venus' looking-glass (*T. biflora*), was first found in Anqing, Anhui Province, in 1981 and subsequently from Zhejiang, Fujian and northern Taiwan in the mid-1980s (Lammers 1998). Both species have strong reproductive ability, and have both 'perfect' and 'cleistogamous' flowers (Chen Lingjing et al. 1992).

Tropic ageratum (*Ageratum conyzoides*)

Originally from Mexico and now widely distributed in tropical areas of the Eastern Hemisphere, tropic ageratum was first recorded by Bentham G. (1861) in the Flora of Hong Kong. The species was subsequently found in southern Yunnan Province in the late 19th century. It is now widespread in lowlands, mountains, hills and plains in the Yangtze Drainage and areas further south. Another species, Mexican ageratum (*A. houstonianum*) from North America was introduced as an ornamental plant in southern China where it has often escaped to the wild.

Ragweed (*Ambrosia* spp.)

Ragweed, including common ragweed (*Ambrosia artemisiifolia*) and giant ragweed (*A. trifida*), is native to North America. Common ragweed was first collected in Hangzhou, Jiangsu Province in 1935. Giant ragweed reportedly invaded northeast China in the 1950s. By 1989, ragweed had expanded from centers in Shenyang, Nanjing, Nanchang and Wuhan to include 12 provinces. Both species effectively utilize large amounts of fertilizer, have high productivity, and regenerate well in dry and infertile soils. Their ability to block sunlight causes reduced crop productivity. Pollen from these species is the principal cause of 'hay fever', which produces symptoms of allergic rhinitis and bronchial asthma. Since 1987, some research progress has been made on biological control with *Zygogramma isuturalis* (Coleoptera: Chrysomelidae) introduced from Canada and the former USSR (Li Xiumei 1997; Wan Fanghao et al. 1993).

Horseweed (*Conyza* spp.)

Horseweed (*Conyza canadensis*), a native of North America, was first collected in Yantai (formerly called Chefoo), Shandong Province, in 1860. Within seven years, it had spread to Ningbo in Zhejiang, Jiu Jiang in Jiangxi, Yichang in Hubei and Nanxi in Sichuan. Today, its range includes all of China. Flax-leaf fleabane (*C. bonariensis*), a related species from South America, was first collected in Hong Kong in 1857 and rapidly spread to Guangdong and Shanghai. A specimen was also collected in Chongqing in 1887. This species and the naturalized Sumatra fleabane (*C. sumatrensis*), also native to South America, are taller than horseweed. These species occur primarily south of the Yangtze River and apparently do not adapt well to the cold, dry climate of northern China.

Daisy fleabane (*Erigeron annuus*)

Daisy fleabane is native to North America. According to Forbes and Hemsely (1890), it was first collected in Shanghai in 1886. It is now found throughout most of China, and is common in both temperate and subtropical regions. A related species from North America, the Philadelphia daisy or Philadelphia fleabane (*E. philadelphicus*), was introduced later to China and now is spreading in Jiangsu, Zhejiang and Shanghai.

Crofton weed (*Eupatorium adenophorum*)

Crofton weed is native to Central America and naturally spread into southern Yunnan from Myanmar along roads in the 1940s. It is now widespread in southwest China, and covers an area of 247,000 km² in Yunnan. This poisonous plant inhibits growth and may even kill local plants and domestic animals (NEPA 1998). Another weed from Central America, fragrant eupatorium (*E. odoratum*), was cultivated in Thailand in the early 1920s (Gagnepain 1924). According to specimen records, this species was present in southern Yunnan in the early 1930s. It is now spreading in Yunnan, Guangxi and Hainan Provinces.

South American climber (*Mikania micrantha*)

South American climber was found in Hainan and south of Guangdong (Zhanjiang, Yangjiang, Taishan, Guangzhou, Zhuhai, Shenzhen and Hong Kong) in the late 1980s where it dominates large areas. It was first introduced into Malaysia from South America and then spread to all of southeast Asia. It climbs trees, blocks sunshine and then kills the trees. Its seeds are spread by wind and could reach even remote areas and islands. For example, Neilingding Island in Shenzhen, Guangdong Province, is famous for its abundant macaques (*Macaca mulatta*). South American climber was first found in 1997 in the Futian national nature reserve located in the Island. After 2 years, it has covered 40–60% of the total land, killed local plants in a large number and threatening over 600 macaques lived in the nature reserve (J. Fellows, pers. comm.; Zhuang Qiushi 1999).

Tall goldenrod (*Solidago altissima*)

Tall goldenrod, a native of North America was introduced from Japan to Taipei in Taiwan in 1935 as an ornamental plant. It was subsequently introduced into Shanghai and Lushan in Jiangxi, and has become wild in these regions. In Shanghai, it is now found in suburbs along the Kunshan-Shanghai railway, and in Pudong and Qingpu. The species reproduces prolifically by seeds and also spreads by rootstocks. It outcompetes other plants and has become a dominant species in many areas. It is a particularly aggressive in disturbed areas, such as suburban wastelands, roadsides, banks, and in residential and industrial areas; and is spreading from such areas into surrounding orchards, croplands, and vegetable fields, especially in Shanghai (Che Jindian and Guo Xihong 1999). Goldenrod (*S. canadensis*) has also been introduced and become naturalized in Shanghai, Wuhan and Lushan. Four additional species in the genus from North America, including the bushy goldenrod (*S. graminifolia*), were introduced into the Lushan Botanic Garden from which they have begun to invade local ecosystems.

Common cordgrass (*Spartina anglica*)

Common cordgrass is a hybrid, originally from western Europe, of smooth (*S. alterniflora*) and marshhay cordgrass (*S. patens*). It was introduced to the coast of Jiangsu Province from Denmark, The Netherlands

and the United Kingdom in 1963 to protect banks from erosion and to improve soils. The species was also used for forage and for paper-making materials. During the next 20 years, the species was widely cultivated north to Jingxi, Niaoning and south to beaches in Guangxi, covering more than 30,000 hm² (Zhi Pu 1985). It has become a predominant species in many of these regions, outcompeting other plants and threatening native biota (NEPA 1998).

Darnel ryegrass (*Lolium temulentum*)

Darnel ryegrass is native to Europe, where it is a common weed in wheat fields and consequently is often found as a contaminant in grain shipments. It was first found in imported wheat from Bulgaria in 1954. By 1957, the species had become established in Heilongjiang Province. By 1961, its range had expanded from several to 45 counties. The species was subsequently found in imported wheat from Australia, the USA, Canada, Argentina, France, Germany, Turkey, Greece, Egypt and Netherlands. In addition, because of improper trade in imported wheat and lack of quarantine for transportation of wheat within China, darnel ryegrass has now invaded crop fields throughout most of China. It is now reported from all provinces and regions of China except Tibet and Taiwan (Li Yanghan 1998). Its seed is sometimes infested by a fungus which makes it poisonous to people and domestic animals (Zhou Ailin and Yue Junshan 1987).

Water hyacinth (*Eichhornia crassipes*)

Water hyacinth is a highly invasive aquatic weed native to South America, which was intentionally introduced into many areas of China as forage for domestic animals, for ornamental use, and to purify wastewater. Since its initial introduction in 1901, the species has spread widely in tropical and subtropical areas, particularly in southern and southwestern China. In 1994, about 10 km² of Dianchi Lake in Yunnan Province – one of China's most famous and beautiful lakes – were completely covered by dense mats of water hyacinth. The rapid spread of this weed has resulted in great economic losses to fisheries and tourism, as well as declines in native aquatic plants and threats to local biodiversity (Ding Jianqing et al. 1995).

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